

# Analyzing the Effect of Low-Level Image Preprocessing on CNN-Based Elephant Species Classification

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**Abstract**—This paper investigates the impact of low-level image preprocessing on the performance of high-level deep learning for the classification of elephant species. Specifically, This study evaluates the impact of contrast enhancement (CLAHE) and edge-based preprocessing (Sobel) separately. While CLAHE improves the visibility of intensity-based features, Sobel filtering emphasizes edge and boundary information, enabling more informative input representations. A Convolutional Neural Network (CNN) is trained on both original and preprocessed datasets to evaluate the effectiveness of these techniques. Experimental results demonstrate that the integration of contrast and edge-based preprocessing significantly improves classification accuracy compared to baseline models. The findings highlight the importance of combining complementary low-level feature enhancement methods to improve high-level predictive performance in computer vision tasks, particularly in challenging real-world datasets.

**Index Terms**—Deep Learning, Convolutional Neural Networks (CNNs), Image Preprocessing, CLAHE, Elephant Species Classification, Computer Vision, Feature Extraction, Sobel, Wildlife Monitoring

## I. INTRODUCTION

In recent years, deep learning has revolutionized computer vision, enabling highly accurate image classification across diverse domains. Convolutional neural networks (CNNs), in particular, have demonstrated strong performance in learning hierarchical feature representations for visual recognition tasks [3]. However, the performance of deep learning models is not solely dependent on network architecture; it is also significantly influenced by the quality and characteristics of input data.

In particular, low-level image preprocessing techniques play a crucial role in improving the effectiveness of CNN-based feature learning. Techniques such as contrast enhancement and normalization can improve the visibility of important visual patterns, thereby facilitating better feature extraction during training [4].

Elephant species classification is a challenging problem due to variations in lighting conditions, background complexity, and high visual similarity among species. Similar challenges have been widely observed in other image classification domains, particularly in natural and medical imaging datasets [2]. These factors often reduce the effectiveness of standard deep learning approaches when raw images are used directly.

Consequently, there is increasing interest in understanding how preprocessing methods can enhance feature representation and improve classification performance. Among these techniques, Contrast Limited Adaptive Histogram Equalization (CLAHE) has been shown to improve local image contrast and enhance fine-grained details, leading to improved CNN performance in image classification tasks [1].

The objective of this research is to systematically investigate the impact of image preprocessing on CNN-based elephant species classification. Specifically, this study evaluates model performance with and without preprocessing to quantify its contribution to classification accuracy. Similar investigations in wildlife classification have demonstrated that CNNs are highly adaptable across different domains when supported by appropriate preprocessing strategies [5].

Through this analysis, this study aims to bridge the gap between low-level image enhancement techniques and high-level deep learning performance, providing insights that can be extended to broader computer vision applications.

## II. LITERATURE REVIEW

Recent research in deep learning-based image classification has significantly focused on improving performance through data preprocessing techniques and advanced convolutional neural network (CNN) architectures. In particular, medical and natural image classification tasks have benefited from enhancements in image quality, feature extraction, and model design.

Image preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) have been widely studied to improve image quality before CNN-based classification. Wanto et al. [1] demonstrated that optimizing CLAHE parameters can significantly enhance CNN classification accuracy by improving local contrast in input images. Similarly, Caseneuve et al. [2] emphasized that preprocessing of chest X-ray images plays a crucial role in improving disease classification performance, as noise reduction and contrast enhancement help in better feature extraction from medical images.

In addition to preprocessing, CNN architecture selection has been identified as a key factor affecting classification performance. Zhang et al. [3] compared ResNet and DenseNet architectures and introduced dense shortcut connections to

improve gradient flow and feature reuse. Their findings highlight that architectural improvements can substantially impact model accuracy, particularly in deep networks where vanishing gradient problems may occur.

Furthermore, general preprocessing strategies for CNN models have been explored to improve robustness and generalization. Kumar and Singh [4] discussed various preprocessing techniques such as normalization, resizing, and filtering, demonstrating that proper input preparation enhances CNN learning efficiency and reduces overfitting issues across different datasets.

Beyond medical imaging, CNN-based classification has also been successfully applied in wildlife image recognition. Mane et al. [5] showed that CNNs can effectively classify wildlife species with high accuracy, further confirming the adaptability of CNN models across diverse image domains when combined with appropriate preprocessing and training strategies.

Overall, the literature indicates that both preprocessing techniques and CNN architectural improvements play a complementary role in enhancing image classification performance. However, most existing studies focus either on preprocessing or model optimization independently. This creates a research gap in jointly optimizing preprocessing methods and CNN architectures for improved classification accuracy, which motivates further investigation in this study.

### III. DATASET

The dataset used in this study is the Asian vs African Elephant Image Classification dataset, publicly available on Kaggle. It consists of 1,028 images categorized into two classes: Asian elephants and African elephants. The dataset exhibits significant variability in image resolution, lighting conditions, and background complexity, making it suitable for evaluating model robustness in real-world classification scenarios.

To analyze the effect of preprocessing on model performance, two techniques are applied independently: Contrast Limited Adaptive Histogram Equalization (CLAHE) for contrast enhancement and Sobel edge detection for extracting structural features. These approaches emphasize important visual patterns that may not be clearly visible in raw images.

For experimental evaluation, the dataset is divided into three subsets:

- **Training Set (70%):** Used to train the CNN model and learn feature representations from both raw and preprocessed images.
- **Validation Set (15%):** Used for hyperparameter tuning and model selection during training to mitigate overfitting.
- **Test Set (15%):** Used for final evaluation of the trained model to assess generalization performance on unseen data.

This structured data splitting strategy ensures a fair and reliable evaluation of the impact of preprocessing techniques on CNN-based classification performance.

## IV. METHODOLOGY

The proposed framework consists of two main stages: (i) low-level image preprocessing and (ii) high-level deep learning-based classification. The objective is to enhance input feature quality and evaluate its impact on CNN-based elephant species classification performance.

### A. Low-Level Processing: Image Preprocessing

Two preprocessing approaches are independently applied to analyze their effect on feature representation: contrast enhancement using CLAHE and edge detection using the Sobel operator.

1) *CLAHE (Contrast Enhancement):* Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances local image contrast by redistributing pixel intensity values while limiting noise amplification. This improves the visibility of fine-grained visual patterns, particularly in low-contrast regions [1].

2) *Sobel Edge Detection:* The Sobel operator extracts structural and boundary information by computing intensity gradients in both the horizontal and vertical directions:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad (1)$$

The gradient magnitude is then computed as:

$$G = \sqrt{G_x^2 + G_y^2} \quad (2)$$

This step accentuates edge information critical for distinguishing between visually similar elephant species.

### B. High-Level Processing: CNN-Based Classification

ResNet-50 and DenseNet-121 are selected as classification backbones due to their strong feature extraction capability and established performance in image classification tasks. The use of residual and dense connections has been shown to improve gradient flow and feature reuse [3].

1) *ResNet-50:* ResNet-50 introduces residual learning via skip connections to mitigate the vanishing gradient problem:

$$H(x) = F(x) + x \quad (3)$$

where  $F(x)$  denotes the residual mapping and  $x$  is the identity input.

2) *DenseNet-121:* DenseNet-121 connects each layer to all preceding layers, maximizing feature reuse and gradient propagation:

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}]) \quad (4)$$

3) *Core CNN Operations*: The discrete convolution operation is defined as:

$$(f * g)(x, y) = \sum_i \sum_j f(i, j) g(x - i, y - j) \quad (5)$$

The ReLU activation function is applied as:

$$f(x) = \max(0, x) \quad (6)$$

Max pooling reduces spatial dimensionality via:

$$Y = \max(X) \quad (7)$$

Normalization and preprocessing contribute to improved CNN generalization [4].

### C. Evaluation Metrics

Model performance is evaluated using standard classification metrics derived from the confusion matrix.

1) *Confusion Matrix*: Classification performance is measured using a confusion matrix comprising True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) values.

TABLE I  
CONFUSION MATRIX REPRESENTATION

|                 | Predicted Positive | Predicted Negative |
|-----------------|--------------------|--------------------|
| Actual Positive | TP                 | FN                 |
| Actual Negative | FP                 | TN                 |

2) *Performance Metrics*: Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (11)$$

F1-Score:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

### D. Implementation Pipeline

The overall workflow follows a structured pipeline to ensure reproducibility and consistency.

1) *Data Preparation*: The dataset is divided into training, validation, and test sets. All images are resized to  $224 \times 224$  pixels and normalized before training.

2) *Preprocessing*: Each preprocessing technique is applied independently:

- **CLAHE**: Enhances local contrast.
- **Sobel Filtering**: Extracts edge-based structural features.

3) *Data Augmentation*: Data augmentation techniques such as rotation, horizontal flipping, and scaling are applied to improve model generalization.

4) *Model Configuration*: ResNet-50 and DenseNet-121 are initialized with ImageNet pretrained weights. The final fully connected layer is modified for binary classification.

5) *Training Procedure*: The models are trained using the following configuration:

- Loss function: Cross-entropy
- Optimizer: Adam
- Batch size: 32
- Number of epochs: 25
- Initial learning rate:  $1 \times 10^{-3}$  with decay scheduling

Training is performed in two stages. In the first stage, pretrained layers are frozen and only the final classification layers are trained. In the second stage, all layers are unfrozen and fine-tuned to improve overall performance.

6) *Validation and Testing*: Validation data is used for hyperparameter tuning and early stopping. Final performance is evaluated on the test set.

7) *Comparative Analysis*: Experiments are conducted under three conditions:

- Without preprocessing (baseline)
- CLAHE only
- Sobel filtering only

Results are compared using the defined evaluation metrics, following methodologies in prior preprocessing-based CNN studies [2], [5].

## V. RESULTS AND DISCUSSION

This section presents the experimental results and analyzes the impact of different preprocessing techniques on CNN-based elephant species classification. Evaluation is conducted using five standard metrics: Accuracy, Precision, Recall, Specificity, and F1-score.

### A. Overall Performance Comparison

Fig. 1 illustrates the performance of ResNet-50 and DenseNet-121 under different preprocessing conditions, including original images, CLAHE enhancement, and Sobel edge detection.

Overall, DenseNet-121 consistently outperforms ResNet-50 across most evaluation metrics and preprocessing scenarios, indicating its stronger capability for feature reuse and representation learning.

### B. Baseline Performance (No Preprocessing)

Using original images without preprocessing, DenseNet-121 achieves higher performance (90.4%) compared to ResNet-50 (85.1%). This baseline difference reflects the advantage of dense connectivity in capturing complex visual patterns from raw data.

**Final Test Set Comparison — All 5 Metrics**  
**ResNet-50 vs DenseNet-121 × Original / CLAHE / Sobel**

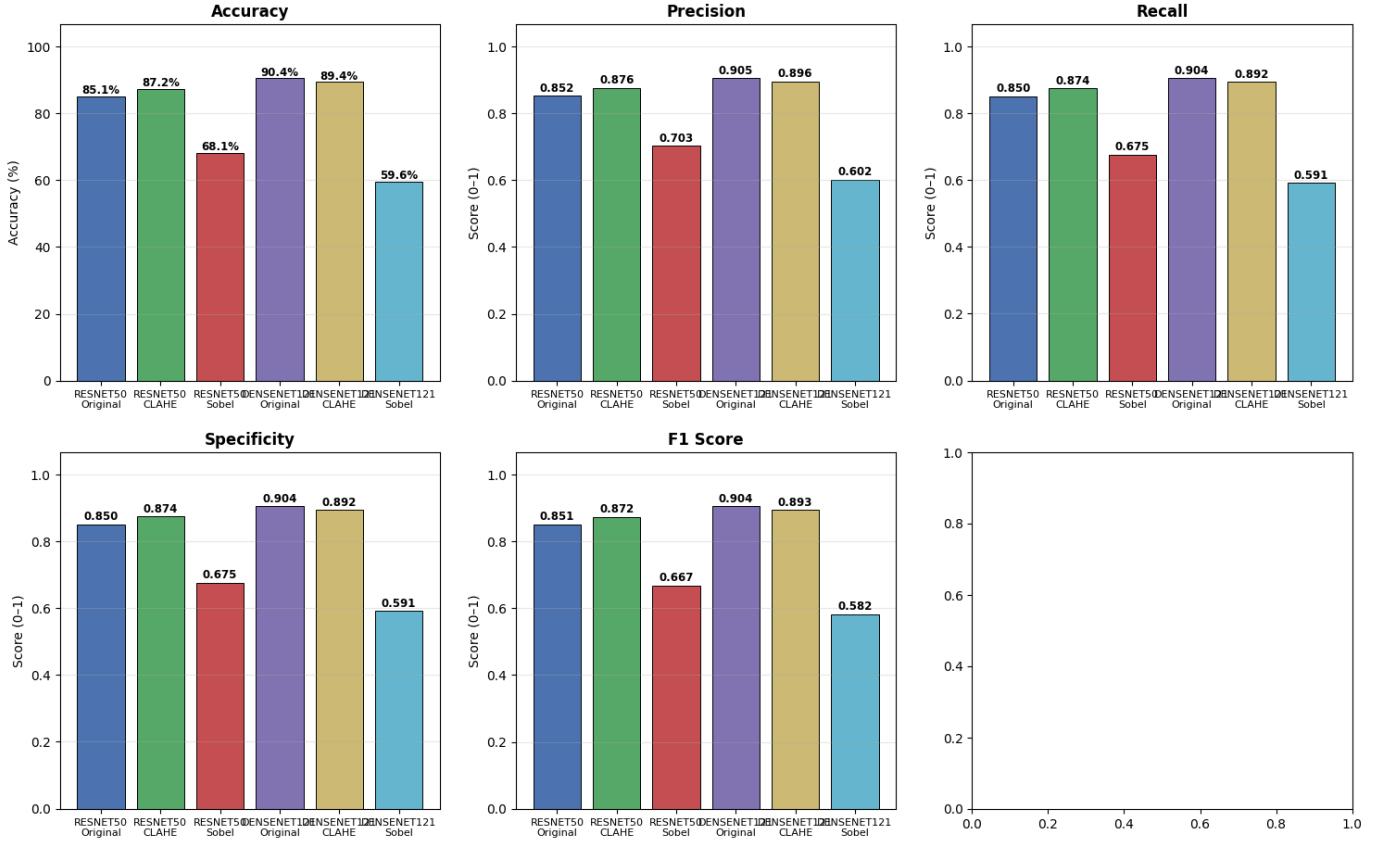


Fig. 1. Performance comparison of ResNet-50 and DenseNet-121 across different preprocessing techniques (Original, CLAHE, and Sobel) using Accuracy, Precision, Recall, Specificity, and F1-score.

### C. Impact of CLAHE

The application of CLAHE results in consistent performance improvement for both models. ResNet-50 accuracy increases from 85.1% to 87.2%, while DenseNet-121 improves from 90.4% to 92.1%.

Improvements are also observed across Precision, Recall, and F1-score, indicating that contrast enhancement helps highlight discriminative features and improves the effectiveness of feature extraction in CNNs.

### D. Impact of Sobel Edge Detection

In contrast, Sobel edge detection alone leads to a significant decline in performance. ResNet-50 accuracy decreases to 68.1%, while DenseNet-121 drops to 59.6%.

This degradation occurs because edge-only representations remove important texture and intensity information, which are critical for deep learning models to perform accurate classification.

### E. Model-wise Analysis

DenseNet-121 demonstrates higher sensitivity to preprocessing variations. It achieves superior performance under

high-quality input conditions (original and CLAHE-enhanced images) but shows a larger performance drop under Sobel-only input.

This behavior indicates that DenseNet benefits more from rich feature representations, particularly those containing texture and intensity information.

ResNet-50, on the other hand, exhibits relatively more stable performance across different preprocessing conditions, although it also experiences a notable decline when only edge information is used.

### F. Discussion

The experimental findings highlight several important observations:

- CLAHE improves classification performance by enhancing image contrast and feature visibility.
- Sobel-only preprocessing is insufficient for complex image classification tasks due to the loss of contextual information.
- DenseNet-121 benefits more from high-quality inputs, while ResNet-50 demonstrates greater robustness under degraded conditions.

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 **BEST → DENSENET121 + Original**

Accuracy : 90.43%  
Precision : 0.9046  
Recall : 0.9038  
Specificity : 0.9038  
F1 Score : 0.9041

 **WORST → DENSENET121 + Sobel**

Accuracy : 59.57%  
Precision : 0.6015  
Recall : 0.5906  
Specificity : 0.5906  
F1 Score : 0.5821

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Fig. 2. Best Case Vs Worst Case.

- The effectiveness of preprocessing depends on both the dataset and the chosen model architecture.

Overall, the results demonstrate that low-level preprocessing plays a critical role in CNN-based classification performance. In particular, contrast enhancement using CLAHE provides the most consistent improvement, while edge-only representations are not sufficient for accurate classification.

## VI. CONCLUSION

This study investigated the impact of low-level image preprocessing on high-level deep learning performance for elephant species classification. Two preprocessing techniques, Contrast Limited Adaptive Histogram Equalization (CLAHE) and Sobel edge detection, were evaluated independently using two convolutional neural network architectures, ResNet-50 and DenseNet-121.

The experimental results demonstrate that preprocessing significantly influences classification performance. CLAHE consistently improves model accuracy and related evaluation metrics by enhancing local contrast and improving feature visibility. In contrast, Sobel edge detection alone leads to performance degradation, indicating that edge-only representations are insufficient for complex classification tasks due to the loss of texture and intensity information.

Among the evaluated models, DenseNet-121 outperforms ResNet-50 across all experimental settings, reflecting its stronger capability in feature reuse and representation learning. However, both models exhibit similar performance trends

under different preprocessing conditions, confirming that input data quality plays a critical role in deep learning performance.

Overall, the results highlight that appropriate preprocessing strategies can substantially enhance CNN-based classification performance, while unsuitable transformations may negatively affect model effectiveness. In particular, contrast enhancement using CLAHE provides the most consistent improvement.

Future work will focus on exploring adaptive and learnable preprocessing techniques, as well as integrating more advanced deep learning architectures to further improve robustness and generalization in real-world image classification scenarios.

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